

DARSHAN THAKER

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RESEARCH INTERESTS

I develop inference-time algorithms that steer pretrained generative models towards desired objectives. My work spans three directions:

- **Language Model Steering:** Efficient reward-model guidance and activation-space steering.
- **Diffusion Model Steering:** Posterior sampling algorithms for image restoration and medical imaging.
- **LLM Robustness:** Optimization-based hallucination elicitation and jailbreak attacks.

EDUCATION

University of Pennsylvania

Sep 2020 – Dec 2026 (expected)

Ph.D., Computer Science.

Columbia University

2018 – 2019

M.S., Computer Science.

The University of Texas at Austin

2014 – 2018

B.S., Computer Science and Pure Mathematics.

EXPERIENCE

University of Pennsylvania

Sep 2020 – Present

Ph.D. Researcher. Advisor: Prof. René Vidal.

Inference-Time Alignment and Steering of Language Models ([Preprint 2026](#); [NeurIPS 2024](#))

- Developed FastGuide, a decoding algorithm that makes diffusion LLM guidance with reward models up to 4.4 $\times$ faster while maintaining generation quality, using KV caching and sparse attention recomputation.
- Co-designed PaCE, an inference-time framework that uses sparse coding to identify and steer concepts in LLM activation spaces, controlling model behavior without retraining.

Red-Teaming and Hallucination Elicitation for LLMs ([ICML 2026](#); [NeurIPS 2025](#); [Preprint 2025](#))

- Mentored a junior PhD student in developing optimization-based methods (SECA, REALISTA) that elicit hallucinations from LLMs.
- Contributed to KDA, a distilled attacker that generates diverse jailbreak prompts for LLMs.

Controllable Generation with Diffusion Models ([ICCV 2025](#); [Preprint 2026](#))

- Introduced Frequency-Guided Posterior Sampling (FGPS), a training-free method that uses timestep-dependent frequency uncertainty information to steer diffusion models to restore degraded images.
- Developed ANaLOG, a probabilistic framework to model uncertainty in latent diffusion models for solving ill-posed image inverse problems.

Defenses for Adversarial Attacks in Computer Vision ([ICML 2022](#); [Preprint 2023](#); [ICLR-W 2026](#))

- Developed sparse coding algorithms that recover the deception type for adversarially perturbed images.
- Co-developed a framework extension to defend vision-language models against semantic attacks.

Siemens Healthineers

Summer 2025

Research Intern. Mentor: Dr. Mariappan Nadar.

- Trained a scale-cascaded diffusion model from scratch on a multi-GPU cluster and developed a posterior-sampling algorithm to steer it, applied to medical image super-resolution ([ISBI 2026](#), *oral*).

Salesforce Research

Spring 2020

Research Intern. Mentor: Dr. Yu Bai.

- Analyzed theoretical conditions under which deep networks learn transferable features, focusing on lower bounds on generalization error.

Earlier Industry Experience

The Curious AI Company	Research Intern, Industrial Process Control	Summer 2018
Facebook	Software Engineering Intern, Search Ranking	Summer 2017
Google	Software Engineering Intern, Location Segmentation	Summer 2016
Symantec	Machine Learning Intern, Center for Advanced ML	Summer 2015

SELECTED PUBLICATIONS

· D. Thaker , L. MacDonald, R. Vidal. <i>FastGuide: Accelerating Reward Guidance for Diffusion Large Language Models</i> . arXiv:2609.36202	Preprint 2026
· D. Thaker , L. MacDonald, R. Vidal. <i>ANaLOG: Anisotropic Native-Latent Operator Guidance for Solving Inverse Problems</i> . arXiv:2609.31933	Preprint 2026
· B. Liang, J. Luo, L. Peng, K. Chan, D. Thaker , K. A. Kinfu, F. Tian, H. Hassani, R. Vidal. <i>REALISTA: Realistic Latent Adversarial Attacks that Elicit LLM Hallucinations</i> . arXiv:2605.12813	ICML 2026
· N. Nguyen, D. Thaker , K. Emmanouilidis, T. Ding, R. Vidal. <i>SSCD: Sparse Semantic Concept Defense Against Semantic Adversarial Attacks to Image Classifiers</i> .	ICLR-W 2026
· D. Thaker , M. Mostapha, R. Miron, S. Qiu, M. Nadar. <i>Scale-Cascaded Diffusion Models for Super-Resolution in Medical Imaging</i> . arXiv:2601.23201	ISBI 2026 (oral)
· B. Liang, L. Peng, J. Luo, D. Thaker , K. Chan, R. Vidal. <i>SECA: Semantically Equivalent & Coherent Attacks for Eliciting LLM Hallucinations</i> . arXiv:2510.04398	NeurIPS 2025
· D. Thaker , A. Goyal, R. Vidal. <i>Frequency Guided Posterior Sampling for Diffusion-Based Image Restoration</i> . arXiv:2411.15295	ICCV 2025
· B. Liang, K. Chan, D. Thaker , J. Luo, R. Vidal. <i>KDA: A Knowledge-Distilled Attacker for Generating Diverse Prompts to Jailbreak LLMs</i> . arXiv:2502.05223	Preprint 2025
· J. Luo, T. Ding, K. Chan, D. Thaker , A. Chattopadhyay, C. Callison-Burch, R. Vidal. <i>PaCE: Parsimonious Concept Engineering for Large Language Models</i> . arXiv:2406.04331	NeurIPS 2024
· D. Thaker , P. Giampouras, R. Vidal. <i>A Linearly Convergent GAN Inversion-based Algorithm for Reverse Engineering of Deceptions</i> . arXiv:2306.04756	CPAL 2024
· D. Thaker* , P. Giampouras*, R. Vidal. <i>Reverse Engineering ℓ_p Attacks: A Block-Sparse Optimization Approach with Recovery Guarantees</i> . arXiv:2203.04886	ICML 2022

TEACHING

Teaching Assistant, University of Pennsylvania <i>Deep Generative Models (Prof. René Vidal).</i>	Fall 2023/Fall 2024
Course Assistant, Columbia University <i>Deep Learning (Prof. Iddo Drori), Analysis of Algorithms (Prof. Alexandr Andoni).</i>	2019

HONORS AND AWARDS

ICCV Doctoral Consortium	2025
AWS-AI ASSET Fellowship, University of Pennsylvania	2024
Course Assistant Fellowship (full tuition waiver), Columbia University	2019
Turing Scholars Honors, The University of Texas at Austin	2018
University Honors, The University of Texas at Austin	2014–2018

SKILLS

Programming	Python, Bash, C++
ML	PyTorch, TensorFlow, Hugging Face Transformers, Diffusers
Distributed GPU Computing	PyTorch Distributed, Triton
Infrastructure	SLURM, AWS